Theory Of Computation-CSA1303

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GitHub Link:-https://github.com/SUMANTH111111/CSA1303

Experiment -1:-

Code:-

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[100];
    int state = 0;
    printf("Enter a string over {a, b}: ");
    scanf("%s", str);
    for (int i = 0; str[i] != '\0'; i++) {
        switch (state) {
            case 0: // q0 - initial state
                if (str[i] == 'a')
                    state = 1;
                else if (str[i] == 'b')
                    state = 3;
                else {
                    printf("Invalid input! Use only a and b.\n");
                    return 0;
                }
                break;
            case 1: // q1 - ends with a
                if (str[i] == 'a')
                    state = 1;
                else if (str[i] == 'b')
                    state = 2;
                break;
            case 2: // q2 - ends with b
```

```
            if (str[i] == 'a')
                state = 1;
            else if (str[i] == 'b')
                state = 2;
            break;
        case 3: // qd - dead state
            state = 3;
            break;
    }
  }
  if (state == 1)
    printf("String Accepted\n");
  else
    printf("String Rejected\n");
  return 0;
}
```

OutPut:-

